

CAMELS-CH – glaciers neighboring countries

For catchments in neighboring countries only the temporal development of glacier area is available based on the glacier inventories of the years 2003 and 2015 from Paul et al. (2011 & 2020), which cover the entire Alpine region. Note that there are several national glacier inventories such as for France in 2006-2009 (Gardent et al. 2014) and Austria in 2012 (Fischer et al. 2015). However, to provide the best possible temporal and methodological consistency, the two glacier inventories by Paul et al. (2011 & 2020) were selected. The glacier inventory from 2003 is based on Landsat (Thematic Mapper, 30m) and the one from 2015 is based on Sentinel-2 (S2, 10m) satellite data. For the latter, former national glacier inventories have been used additionally as guidance to map the glacier outlines (Paul et al. 2011 & 2020).

The more recent glacier inventory from Paul et al. (2020) seems to be more accurate than the inventory from 2003 (Paul et al. 2011) probably due to the more precise data and more experience in mapping glacier outlines. This means that certain glaciers present a larger area in 2015 than in 2003 and appear to have grown falsely over the years. To correct this error, the area development rate of Swiss glaciers of similar area in the year 2015 was applied to these glaciers and then used to calculate their temporal area development from 1980 to 2021.

References

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Gardent, M., Rabatel, A., Dedieu, J.P. & Deline, P. (2014) : Multitemporal glacier inventory of the French Alps from the late 1960s to the late 2000s. *Global and Planetary Change*, 120: 24-37. dx.doi.org/10.1016/j.gloplacha.2014.05.004

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Paul, F., Rastner, P., Azzoni, R.S., Diolaiuti, G., Fugazza, D., Le Bris, R., Nemec, J., Rabatel, A., Ramusovic, M., Schwaizer, G. & Smiraglia, C. (2020): Glacier shrinkage in the Alps continues unabated as revealed by a new glacier inventory from Sentinel-2. *Earth System Science Data*, 12: 1805-1821. doi.org/10.5194/essd-12-1805-2020

Instructions

Attributes:

- name, year_acq, year_rel, area_km2, length_km, masl_min, masl_mean, masl_max, slope_deg, aspect_deg, GLIMS_ID, CAMELS_ID

name	= name of the glacier
year_acq	= year when the glacier outlines were mapped
year_rel	= year when the glacier inventory was released
area_km2	= area of glacier [km ²]
length_km	= length of the glacier [km]
masl_min	= minimum elevation of glacier [m a.s.l.]
masl_mean	= average elevation of glacier [m a.s.l.]
masl_max	= maximum elevation of glacier [m a.s.l.]
slope_deg	= slope of the glacier in degrees
aspect_deg	= aspect of the glacier in degrees
GLIMS_ID	= international glacier ID from the GLIMS database
CAMELS_ID	= newly created glacier ID specifically for CAMELS-CH

Source data:

- 2015 glacier inventory Paul et al. (2020): [Paul, F et al. \(2019\): Glacier inventory of the Alps from Sentinel-2, shape files \(pangaea.de\)](#)
- 2003 glacier inventory Paul et al. (2011): <https://www.glims.org/maps/glims>

Instructions:

1. Get the data and open shapefiles from the glacier inventories (GIs) in QGIS/ArcGIS
2. Extract the glaciers from the neighboring countries which still belong to the CAMELS-CH catchments (France, Italy and Austria)
3. Export the selected glaciers as .csv from QGIS/ArcGIS and join the two glacier inventories and determine which glacier from GI 2003 belongs to which glacier from GI 2015
4. Since the two glacier inventories do not have a common glacier ID, create a new ID indicating the country, year of glacier inventory and number of glacier. For example:
 - FR03_01 & FR15_01 (FR = France, 03 = GI from 2003, 15 = GI from 2015, 01 = first glacier)
 - If over the years, one complete glacier in 2003 broke up into several pieces in 2015 → FR03_01 = FR15_01a, FR15_01b
 - If in 2003 several small glaciers were mapped and in 2015 it is one big glacier → FR03_01a, FR03_01b = FR14_01
 - For calculating the temporal development of the glacier area in a later step, sum the area of the glaciers FR15_01a and FR15_01b to one glacier (FR15_01)
5. Determine the glaciers that have a bigger area in 2015 than in 2003 or missing glaciers in 2003/2015
6. Extract from the provided area development time series of Swiss glaciers the area development rate of glaciers, which have a similar area in the year 2015 (or the missing year) and apply this rate to the glaciers of the neighboring countries.
7. Having the areas for the years 2003 and 2015 for all glaciers, calculate the area for each year from 1980 to 2021 (linear interpolation → $y = y_1 + (y_2 - y_1) * ((x - x_1) / (x_2 - x_1))$)
8. Output = shapefiles of both inventories (see attributes above) and time series of the area development of the glaciers